

Summary

This is Iris, a Software Engineering Ph.D. candidate at the University of California, Irvine (UCI), working with Professor **Cristina Videira Lopes**. My primary research area is Software Engineering, focused on **AI for Reliable Software Engineering**. Currently working on LLM memorization, hallucination detection, retrieval-augmented generation (RAG), verified program synthesis with Dafny, and query-adaptive fusion methods for production search ranking.

Education

 University of California, Irvine <i>Ph.D. in Software Engineering</i>	Irvine, California Sep. 2022 – Dec. 2027
 University of California, Irvine <i>M.S. in Software Engineering</i>	Irvine, California Sep. 2022 – Jun. 2026
 Iowa State University <i>B.S. in Computer Science, and B.S. in Agronomy</i>	Ames, Iowa Sep. 2017 – May 2022

Technical Skills

Languages: Python, Java, C, C++, C#, HTML/CSS, JavaScript, SQL

Developer Tools: VS Code, Eclipse, Google Cloud, Android Studio, AWS, Azure

AI/ML&DL: LLM fine-tuning, RAG, Prompting, AI Agent Dev, LLM Hallucination Detection, RL

Frameworks: LangChain, HuggingFace, Unsloth, Linux, Scikit-learn, PyTorch, TensorFlow, Git

Publications

[Under Review]	Optimal Fusion Weights Form Intervals, Not Points: A Latency-Aware Study of Query-Adaptive WRRF. Iris Ma , Zahra Shekarchi, Stuart Mcilroy, Mahtab Farrokh, Fattane Zarrinkalam, and Joel Stremmel
[Under Review]	VeriAct: Beyond Verifiability – Agentic Synthesis of Correct and Complete Formal Specifications. Md Rakib Hossain Misu, Iris Ma , and Cristina V. Lopes
[ACL L2M2 2025]	Memorization: A Close Look at Books. Iris Ma , Ian Domingo, Alberto Krone-Martins, Pierre Baldi, and Cristina V. Lopes
[IPLDSC 2025]	Development of a CDH-Specific Conversational Assistant Using RAG. Vanessa Klotzman, Cristina V. Lopes, John Schomberg, Yostina S. Armanyous, Danielle Linden, Iris Ma , et al.
[SIGCSE 2024]	Integrating AI Tutors in a Programming Course. Iris Ma , Alberto Krone Martins, and Cristina V. Lopes
[FSE 2024]	Towards AI-Assisted Synthesis of Verified Dafny Methods. Md Rakib Hossain Misu, Cristina V. Lopes, Iris Ma , and James Noble
[arXiv 2024]	Commit Messages in the Age of Large Language Models. Cristina V. Lopes, Vanessa I. Klotzman, Iris Ma , and Iftekar Ahmed
[ICSE SEENG 2023]	Improving the Quality of Commit Messages in Students' Projects. Iris Ma , and Cristina V. Lopes

Work Experience

 University of California, Irvine <i>Research Assistant</i>	Irvine, California September 2022 – Present
- Developed RAGMan, an RAG-based AI tutor used by 1,000+ UCI students and CHOC doctors; secured \$48K in funding. - Served as TA for Information Retrieval, teaching 1,000+ students to build tokenizers and search engines. - Investigated LLM training-data memorization and membership inference by extracting full books from Llama. - Fine-tuned Llama via LoRA with greedy decoding, showing SFT can undo alignment's memorization mitigations. - Researched LLM proficiency in verified programming with Dafny, contributing verified solutions to benchmark datasets. - Developed hallucination detection methods for LLMs to improve reliability and accuracy. - Evaluated LLMs for commit message generation, showing they outperform prior methods in accuracy and quality. - Improved student commit message quality by 66% by modifying GitHub Desktop to provide structured, guided prompts.	
 Thomson Reuters <i>Applied Scientist Intern</i>	Remote June 2026 – September 2026
- Designed a document-enrichment and query-expansion technique, improving retrieval nDCG@10 by up to 68% over BM25. - Built an LLM-based topic-mining pipeline labeling 50K+ documents to enable category-aware retrieval. - Standardized latency-vs-quality Pareto evaluation with significance testing to keep methods deployable in production. - Built an LLM relevance-judging pipeline to fill sparse qrels, adding 8,900+ verified judgments at 98.9% precision to correct benchmark recall ceilings and enable fairer method comparison.	

Thomson Reuters

Applied Scientist Intern

Eagan, Minnesota

June 2025 – September 2025

- Built 10 query-adaptive fusion methods (WRRF) integrated into production search, beating standard RRF.
- Discovered optimal fusion weights form intervals, not points, explaining RRF's near-optimality for 10–38% of queries.
- Designed an oracle/headroom-recovery framework with latency-aware tiering for production trade-offs.
- Engineered four sparse-dense retriever configs (BM25/RM3, MiniLM, Qwen3); authored an arXiv paper on the findings.

Corteva Agriscience

Software Engineer Intern

Johnston, Iowa

June 2023 – September 2023

- Led a team of 7 building an image-analysis web app with Angular, .NET, Entity Framework, and Azure.
- Improved data collection efficiency by 90% and reduced ergonomic injury risk from manual measurement.
- Built an Azure trigger function enabling real-time, unlimited-upload monitoring of new images.
- Deployed an object-detection microservice (80%+ accuracy) with cloud API integration for image retrieval.

Iowa State University

Research Assistant

Ames, Iowa

May 2021 – May 2022

- Enhanced a compiler to support the latest Java language features, improving performance and capabilities.
- Researched adoption of new Java language features in open-source GitHub projects using Python and Excel.

Presentations & Talks

-  “Large Language Models (LLMs) on a Daily Basis”, The Pacific Club [Newport Beach, California, USA, 2025]
-  “Improving the Quality of Commit Messages in Students’ Projects”, ICSE SEENG [Melbourne, Australia, 2023]
-  “New Java Language Features Adoption on GitHub Hosted Open Source Projects”, Undergraduate Research Symposium [Ames, Iowa, USA 2022]

Community Service

-  Reviewer, ACM Virtual Global Computing Education Conference (SIGCSE 2024), reviewed 5 papers
-  Student volunteer, Southern California Software Engineering Symposium (SuCSES 2023)
-  Student volunteer, ACM SIGPLAN International Conference on Systems, Programming, Languages and Applications: Software for Humanity (SPLASH 2021)

Scholarships, Awards, & Grants

University of California Irvine

-  [2024] **Beall Applied Innovation’s Graduate Innovation Fellowship** (\$5,000)
-  [2024] **I-Corps Program** (\$5,000)
-  [2024] **Beall Applied Innovation’s Proof-of-Product Grant** (\$48,000)
-  [2023] **ESEC/FSE Conference Travel Fund** (\$2450)
-  [2022] **New Ph.D. Student Fellowship** (\$2500)

Iowa State University

-  [2021] **Robert and Cecile Mitchell Scholarship** (\$2650)
-  [2021] **International Incentive Scholarship** (\$1500)
-  [2021] **Global Scholars Award for Undergraduate Retention** (\$500)
-  [2020] **International Incentive Scholarship** (\$1500)
-  [2018 – 2022] **Dean’s List Honors** (10 times)

Mentoring

-  **Shengdong Jiang**, Undergraduate Student (UWaterloo) (Summer 2026 – present)
-  **Derek Xu**, Undergraduate Student (UCI) (Winter 2024 – Spring 2025)
-  **Jianfeng Ma**, Undergraduate Student (UCI) (Winter 2024 – Spring 2025)
-  **Custo Yang**, Undergraduate Student (UCI) (Winter 2024 – Spring 2025)
-  **Avery Zhuang**, Undergraduate Student (UCI) (Summer 2024)
-  **Farhan Zaman**, Undergraduate Student (UCI) (Summer 2024)
-  **Andy Anh-Huy Tonthat**, Undergraduate Student (UCI) (Summer 2024)
-  **Mohammad Mirzaei**, Undergraduate Student (UCI) (Summer 2024)
-  **Ryan Zhuang**, Undergraduate Student (UCI) (Summer 2024)

References

Dr. Cristina Videira Lopes

Chancellor’s Professor

Donald Bren School of Information and Computer Sciences
University of California, Irvine
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Dr. Alberto Krone-Martins

Assistant Professor

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